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Excited-State Proton Transfer Reaction in a New Benzofuryl 3-Hydroxychromone Derivative: The Influence of Low-Polar Solvents

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Unique properties of a new 3-hydroxychromone derivative 2-(6-diethylamino-benzo[*b*]furan-2-yl)-3-hydroxychromone (FA) in its ability to exhibit excited-state intramolecular proton transfer (ESIPT) reaction are described. In contrast to all other chromone and flavone derivatives studied, in low-polar solvents it exhibits in emission, together with the tautomer (T*) band, an intensive band of the normal (N*) form. Previously the intensive N* form in emission was observed only in highly polar, mostly protic solvents. While its absorption spectra are sensitive to H-bond acceptor properties of the solvent, the fluorescence spectra are not. This suggests that intermolecular H-bonds with a solvent, if they exist in the ground state, are reorganized in the excited state in favor of intramolecular bond, which is the pathway for ESIPT reaction. The energy difference, $\nu_{N^*} - \nu_{T^*}$, between N* and T* emission maxima is in almost ideal correlation with the Reichardt solvatochromic parameter E_N^T. This suggests the use of FA as a highly sensitive polarity sensor. A good correlation between $\nu_{N^*} - \nu_{T^*}$, and the ratio of the N* and T* band intensities is observed. This allows to observe the solvent effects on a manifold increased level of sensitivity. The analysis based on Lippert and Bakhshiev equations and the quantum-chemical calculations suggest a substantial increase of the dipole moment on electronic excitation to the N* state. The appearance of the N* form in emission may be the result of its dielectric stabilization.